



T(6th Sm.)-Computer Science-G/SEC-B-X-2/CBCS

2021

COMPUTER SCIENCE — GENERAL

Paper : SEC-B-X-2

(Information Security)

Full Marks : 80

The figures in the margin indicate full marks.

*Candidates are required to give their answers in their own words
as far as practicable.*

Group - A

1. Answer **any five** questions :

2×5

- (a) What are three independent dimensions in a Cryptographic system?
- (b) What is Brute force attack?
- (c) Write the main differences between block and stream cipher.
- (d) Briefly define a ring.
- (e) Find the value of $321 \bmod 11$ using Fermat's theorem.
- (f) What is the difference between diffusion and confusion?
- (g) What is Secure Electronic Transaction (SET)?
- (h) What is the OSI security architecture?

Group - B

Answer **any four** questions.

5×4

- 2. Draw a simplified model of Symmetric Encryption technique and explain it in brief.
- 3. Give example each for substitution and transposition ciphers.
- 4. Explain Fermat's theorem with suitable example.
- 5. What is the purpose of Digital Signature? How does it provide additional security?
- 6. What is the purpose of P-box in DES?
- 7. Write about the different principles of Security.

Please Turn Over

Group - C

Answer *any five* questions.

8. (a) Write a short note on different types of Wireless Network threats.
(b) What is PGP with respect to Electronic mail security? 5+5
9. (a) What is Anti-Replay Service? Why it is needed?
(b) Write the different benefits of IPSec. 4+6
10. (a) What is IP sniffing and IP spoofing?
(b) Explain Diffie-Hellman key exchange algorithm with example. 4+6
11. (a) Explain substitution technique with suitable example.
(b) What are the principles of Public-Key Cryptosystems? 5+5
12. (a) What is Message digest? Why is it used?
(b) What are the roles of the public and private key? 5+5
13. Write short notes on (*any two*) : 5×2
(a) Use of Public-Key Certificates
(b) Network Access Control (NAC)
(c) Transposition Techniques
(d) RSA Algorithm.
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